

AL-JAWF KUFRA, Libyan Arab Jamahiriya

WMO: 622710

Lat: 24.1986N

Lon: 23.3000E

Elev: 1430

StdP: 13.95

Time Zone: 2.00 (E02)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	43.2	45.4	10.3	9.8	66.4	13.0	11.2	68.1	21.8	61.0	19.7	61.8	4.3	50	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	107.6	65.4	105.1	65.2	103.2	64.7	67.9	100.5	66.7	99.8	65.9	99.1	8.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.8	65.0	82.9	51.8	60.2	83.0	50.2	56.7	84.1	32.8	100.4	31.9	99.7	31.2	99.3	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	18.0	15.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.9	57.6	62.3	70.0	80.0	85.8	90.3	90.6	90.0	86.5	79.3	69.5	59.9
	DBStd	12.99	5.32	6.27	7.19	6.01	6.05	4.18	2.85	2.65	4.13	5.40	5.31	4.68
	HDD50	2	1	1	0	0	0	0	0	0	0	0	0	0
	HDD65	576	239	116	29	0	0	0	0	0	0	0	18	174
	CDD50	9815	237	346	620	899	1109	1210	1260	1241	1095	907	584	307
	CDD65	4914	10	41	184	449	644	760	795	776	645	442	152	16
	CDH74	70239	130	530	2023	5538	9342	12156	12787	12126	9235	4933	1282	157
	CDH80	42459	18	195	936	3010	5792	8056	8377	7732	5468	2426	419	30
Wind	WSAvg	7.7	6.3	6.8	8.1	9.4	8.1	8.4	8.6	8.4	7.7	7.6	6.5	6.1
Precipitation	PrecAvg	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecMax	0.7	0.4	0.3	0.2	0.3	0.1	0.0	0.0	0.0	0.0	0.1	0.3	0.3
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.2	92.6	100.3	105.0	111.0	112.9	107.8	105.4	105.1	102.1	92.2	83.4
		MCWB	54.4	57.2	61.3	63.3	65.6	66.8	66.2	66.2	66.5	64.1	59.2	55.5
	2%	DB	77.6	86.1	94.6	101.1	107.4	108.2	105.5	103.5	103.1	97.9	88.3	78.4
		MCWB	52.1	54.1	58.6	61.7	65.0	65.5	66.3	65.6	65.5	63.5	58.3	53.8
	5%	DB	73.9	80.5	89.9	97.8	103.9	104.9	102.8	101.6	100.7	94.5	85.0	73.8
		MCWB	50.9	53.3	56.7	60.5	63.8	64.6	65.4	65.1	64.7	62.5	57.7	52.4
	10%	DB	69.8	75.5	85.0	94.2	100.2	102.3	100.8	99.9	97.9	91.4	81.7	70.0
		MCWB	49.4	51.0	54.9	59.4	62.8	64.0	64.9	64.7	64.0	61.5	56.9	51.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.6	59.5	62.3	64.9	66.5	67.9	70.5	68.8	68.5	66.8	62.6	56.9
		MCDB	78.5	89.1	97.6	98.2	103.8	107.1	101.1	100.3	100.0	91.2	82.6	78.0
	2%	WB	53.2	55.8	59.1	62.7	65.5	66.5	68.2	67.2	66.7	64.7	60.5	54.9
		MCDB	74.9	81.5	92.2	98.0	103.8	104.8	100.2	99.0	98.6	92.7	81.8	74.1
	5%	WB	51.7	53.5	57.2	61.2	64.4	65.4	66.6	66.2	65.5	63.7	59.1	53.5
		MCDB	69.8	78.4	88.7	94.7	103.3	102.9	99.1	98.2	96.9	91.5	81.4	71.0
	10%	WB	50.3	51.7	55.4	59.7	63.0	64.4	65.6	65.4	64.6	62.5	57.7	52.1
		MCDB	67.3	73.0	83.9	92.5	98.4	101.1	98.3	97.4	95.5	89.1	79.2	68.2
Mean Daily Temperature Range	5% DB	MDBR	21.1	23.3	23.5	24.0	23.4	23.0	20.3	19.5	20.7	20.8	21.5	20.2
		MCDBR	26.5	31.4	30.4	28.2	27.4	26.0	22.1	20.9	23.6	23.7	25.0	24.9
		MCWBR	12.3	14.3	13.4	11.3	10.8	9.7	7.9	7.2	7.8	8.0	10.2	10.8
	5% WB	MCDBR	23.7	28.5	28.1	25.8	25.6	24.3	20.5	19.7	21.5	21.7	22.3	21.8
		MCWBR	11.2	13.2	12.7	10.7	10.3	9.0	7.8	7.2	7.5	7.7	9.4	9.7
Clear-Sky Solar Irradiance	taub		0.362	0.414	0.473	0.593	0.613	0.531	0.503	0.496	0.482	0.440	0.376	0.362
	taud		2.112	1.934	1.784	1.560	1.540	1.722	1.797	1.821	1.849	1.934	2.088	2.109
	Ebn at Noon		281	272	262	235	229	247	254	256	256	261	273	275
	Edh at Noon		44	57	70	89	91	75	69	68	64	56	45	43
All-Sky Solar Radiation	RadAvg		1516	1771	2097	2320	2445	2545	2522	2403	2187	1899	1588	1420
	RadStd		43	80	93	64	69	60	32	36	34	36	33	45

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page